

NIFTY Post-Fall Recovery Study

1. Hypothesis and Research Design

Hypothesis: After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days.

Event Definition

A significant fall is defined as a **daily NIFTY return of $\leq -3.0\%$** .

$$\text{Daily Return} = (\text{Close}_t / \text{Close}_{t-1} - 1) \times 100$$

Recovery Definition

Recovery is measured using forward returns after the event over:

- **1 trading day**
- **3 trading days**
- **5 trading days**

A positive return means NIFTY increased after the entry. However, event returns are also compared with normal non-event days to check whether the recovery is stronger than normal market movement.

Entry and Exit

To avoid look-ahead bias:

- **Entry:** Open of the next trading day
- **Exit:** Close after 1, 3, or 5 trading days

The event-day close is not used as the entry price because it is known only after that trading day ends.

Data

The NIFTY dataset covers **22 September 2015 to 18 September 2026** and contains **2,722 observations** with Date, Open, High, Low and Close.

Data was checked for duplicate dates, date ordering, missing/invalid OHLC values, suspicious observations and actual date coverage.

2. Event Results

The study identified **41 events** under the -3% event definition.

	Horizon Mean	Median	Win Rate	Std. Dev.
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1D	-0.0219%	0.1484%	56.10%	1.7383%
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3D	-0.0398%	0.5508%	58.54%	3.0618%
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	Horizon	Mean	Median	Win Rate	Std. Dev.
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	5D	-0.1129%	0.1856%	51.22%	4.3273%
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The median return is positive for all three horizons, but the mean is slightly negative. This shows that some large negative returns affect the average. Therefore, win rate alone is not enough to prove a recovery effect.

3. Baseline and Statistical Evidence

Event returns were compared with forward returns on normal, non-event NIFTY days.

	Horizon	Event Mean	Baseline Mean	Difference
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	1D	-0.0219%	-0.0695%	+0.0476%
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	3D	-0.0398%	+0.0247%	-0.0645%
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	5D	-0.1129%	+0.1228%	-0.2357%
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The differences are small compared with the volatility of NIFTY returns.

95% Confidence Intervals

	Horizon	95% CI
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	1D	-0.5706% to +0.5268%
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	3D	-1.0062% to +0.9266%
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	5D	-1.4788% to +1.2529%
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All three confidence intervals include zero, so the estimated average event return has substantial uncertainty.

Bootstrap: Event – Baseline

	Horizon	Observed Difference	Bootstrap 95% CI
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	1D	+0.0476%	-0.5115% to +0.5606%
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	3D	-0.0645%	-1.0159% to +0.8208%
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	5D	-0.2357%	-1.6371% to +1.0046%
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All bootstrap intervals also include zero.

The hypothesis-test results were:

	Horizon	t-statistic	p-value
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	1D	0.175	0.431
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Horizon t-statistic p-value

3D -0.1346 0.5532

5D -0.3482 0.6352

These results do not provide statistically significant evidence of a positive event effect at conventional significance levels.

4. Robustness, OOS and Event Dependence

The event threshold was tested from **-2% to -4%**.

Threshold Events Mean 1D Mean 3D Mean 5D

-2.0% 63 +0.229% +0.082% -0.097%

-2.5% 33 +0.218% +0.099% -0.307%

-3.0% 21 +0.552% +0.254% -0.181%

-3.5% 14 +0.554% -0.278% -1.416%

-4.0% 11 +0.783% -0.741% -1.587%

Results change across thresholds and holding periods, so the recovery effect is not consistent across specifications. Testing several thresholds and horizons also creates risks of data snooping, multiple testing and overfitting.

Out-of-Sample

The data was split at **10 July 2024**.

- Development events: **19**
- OOS events: **2**

Horizon Development Mean OOS Mean OOS Win Rate

1D +0.588% +0.2079% 100.0%

3D +0.236% +0.4226% 50.0%

5D -0.3663% +1.5776% 50.0%

The OOS results are positive on average, but only **2 OOS events** are available, so they provide limited independent evidence.

Event Clustering

The minimum gap between events was **8 trading days**.

- Within 1 day: **0**

- Within 3 days: **0**
- Within 5 days: **0**
- Within 10 days: **4**

This reduces concerns about very closely occurring events, although nearby events may still be related to the same market episode.

5. Costs, Limitations and Conclusion

A **0.20% round-trip cost and slippage** assumption was used:

- 0.10% transaction cost
- 0.10% slippage

After costs, the reported mean returns were:

- **1D:** -0.2219%
- **3D:** -0.2398%
- **5D:** -0.3129%

Limitations

The main limitations are the relatively small event sample, changing market regimes, possible dependence between events, threshold sensitivity, transaction costs and the very small OOS sample.

The hypothesis would be weakened or rejected if larger independent samples, reasonable alternative event definitions, or future unseen data consistently failed to show recovery relative to the normal-market baseline.

Conclusion

The results do **not provide strong statistical support** for the hypothesis that a $\leq -3\%$ one-day NIFTY fall is followed by a reliable recovery over the next 1–5 trading days.

The event means were close to zero or slightly negative, all 95% confidence intervals included zero, and event-minus-baseline differences were small and statistically uncertain.

This does not mean recovery never happens. It means that, in this sample, the evidence is **not strong enough to establish a reliable standalone post-fall recovery effect**.